

Aakash Yadav

LinkedIn: <https://www.linkedin.com/in/aakash-yadav294/> GitHub: <https://github.com/AakashYadav6>
Email: aakashyadav5013@gmail.com Mobile: +91 9137087676

EDUCATION

- **Thakur College of Engineering and Technology** Mumbai, India
Bachelor of Engineering in Mechanical Engineering; GPA: 9.33/10 *July. 2020 – June 2024*
- **K.J Somaiya College of Science and Commerce** Mumbai, India
HSC; Percentage 71.08 *Aug. 2019 – July 2020*
- **Sungrace English High School and Junior College** Mumbai, India
SSC; Percentage 89.80 *Aug. 2017 – July 2018*

EXPERIENCE

- **Algobotix Pvt Ltd** India
Robotics Engineer *Sep 2024 – Feb 2025*
 - Led **structural design and experimentation**, developing a **slung rod setup** for multiple drones carrying a common payload.
 - Conducted **slung load experiments** in **ArduPilot/PX4**, including simulations and real-world testing.
 - Designed the **slung load system** in **SolidWorks** and migrated the system from **ROS1 to ROS2** for improved communication and performance.
 - Assembled, calibrated, and tested drones for the embedded team, optimizing **stability and performance** through iterative improvements.
- **E-Yantra Robotics Competition** India
Participant *Sep 2023 – Dec 2023*
 - Developed **Cosmo Logistic**, a lunar warehouse system featuring **racks and mobile robots** for autonomous goods transportation.
 - Implemented **Aruco marker detection** using OpenCV library to identify and track boxes within the warehouse environment.
 - Designed autonomous **Ebot mobile robot navigation** using the **Nav2 stack**, leveraging the **Simple API Commander** for multi-waypoint navigation, **SLAM Toolbox** for mapping, and **AMCL** for localization. Enabled docking, undocking, and dynamic waypoint adjustments.
 - Integrated a **UR5 robotic arm** with **MoveIt** for precise **box manipulation** between warehouse racks.
- **Rigbetellabs ROS Mentorship** Mumbai, India
Trainee *Oct 2023 – Dec 2023*
 - Acquired proficiency in **ROS Basics**, **TurtleSim Simulator**, **SLAM**, and **Navigation** through mentorship program.
 - Successfully implemented ROS-based projects utilizing **SLAM** and **Navigation** techniques.
- **Bureau of Indian Standards** Mumbai, India
Research Assistant *Dec 2023 – Feb 2024*
 - Conducted research on a **"thermal stability test"** apparatus, including the development of code utilizing the **OpenCV library** for the detection of lithium paper. Procured necessary hardware components essential for project implementation.
 - Understanding the concept of **Quality Control** in the industry through sample products. Studied and understood various **Indian Standards (IS)** for products, performed tests on sample products to ensure their conformity with IS, and prepared test reports for the samples.

SKILLS

- **Languages:** Python, C++, HTML, CSS, MATLAB
- **Developer Tools:** VS Code, Anaconda, Pycharm, ROS, Gazebo, Rviz, Solidworks, Ansys Workbench, Fusion 360, Microsoft Offices
- **Libraries:** Pandas, Numpy, Scikit-learn, OpenCV, Matplotlib

RESEARCH WORK

Design and Development of Amphibious Unmanned Aerial Vehicle ([Project Document](#))

- Designed an **Amphibious UAV (A-UAV)** with a **coaxial rotor system**, reducing power consumption by **15%**, increasing thrust, and incorporating an **innovative tilt rotor mechanism** for seamless air-to-water transitions with a **streamlined upper body** for aerodynamic efficiency.
- Conducted **structural, stability, and aerodynamic analysis** using **SolidWorks**, calculated **airframe radius and coaxial height** from *Introduction to Multicopter Design and Control* by Quan Quan to ensure **Stable Equilibrium**, and used **ANSYS** for static structural analysis and **SimScale** for aerodynamic evaluation.
- Led **material and electronic component selection**, determining UAV properties such as **weight, thrust requirements, center of mass, and center of buoyancy**, ensuring optimal performance.
- Demonstrated **integration of ESCs, Brushless DC Motors, and Raspberry Pi** for **efficient PWM-based motor control and propulsion**, applying **multicopter design principles** where **two coaxial propellers generate thrust equivalent to 1.6 single rotors** for enhanced efficiency.

Calculation of Thermodynamic Properties of Li-Br Vapor Absorption Refrigeration System ([Project Document](#))

Ongoing

- Developed **MATLAB** code to analyze and calculate thermodynamic properties of a **Li-Br vapor absorption refrigeration system**, ensuring accuracy by comparing results with existing literature and theoretical calculations.

PROJECTS

- **Autonomous Mobile Robot for Navigation and Object Tracking ([GitHub Repository](#))**
 - Developed a **differential drive robot** with **Raspberry Pi 4, Arduino Nano, DC motors**, using **PID** for control and integrating **RP Lidar, depth camera** for **mapping, localization, and tracking**.
 - Simulated in **Gazebo, RViz** for **path planning, obstacle avoidance**, now focusing on **hardware integration**.
 - Technologies: **ROS2, Python, C++, OpenCV, Gazebo**
- **Line Follower Robot ([Project Document](#))**
 - Designed and developed an **Arduino Uno-based** autonomous robot utilizing **IR sensors** for accurate line detection and object avoidance.
 - Integrated **L298N motor driver** to regulate **BO motor speed**, powered by a **7.4V (2S) Li-Po battery** for efficient performance.
 - Built a custom mobile app using **MIT App Inventor** and **Bluetooth HC-05 module**, enabling real-time remote control with latency under 100ms.

CERTIFICATIONS

- **[Wheeled Mobile Robots](#)** - By Prof. Asokan T, Prof. Santhakumar Mohan — IIT Madras, IIT Palakkad
- **[Robotics and Control: Theory and Practice](#)** - By Prof. N. Sukavanam, Prof. M. Felix Orlando — IIT Roorkee
- **[Industrial Robotics: Theories for Implementation](#)** - By Prof. Arun Dayal Udai — IIT-ISM Dhanbad
- **[C++](#)** - Spoken Tutorial IIT Bombay
- **[Advanced Robotics](#)** - By Prof. Ashish Dutta — IIT Kanpur
- **[Computer Vision and Image Processing](#)** - By Prof. M. K. Bhuyan — IIT Guwahati
- **[Automation in Manufacturing](#)** - Prof. Shrikrishna N. Joshi — IIT Guwahati